

Practice Set No. 2

Counting with Overlaps

Each problem is about groups that share something. Notice what happens when groups overlap.

1. Twenty kids are on the playground. Eight of them are on the slide, five are on the swings, and nobody is on both at the same time. How many kids are doing one of those two things? How many are doing neither?

2. Five friends — **Ana, Beto, Carla, Diego, and Elena** — filled out a survey.

Dog-lovers: *Ana, Beto, Carla.*

Cat-lovers: *Ana, Beto, Diego, Elena.*

(a) Who likes **both** dogs and cats?

(b) How many friends like dogs **or** cats (or both)?

(c) Does $3 + 4$ give the answer to (b)? If not, why not?

3. In a club of 10 kids, 6 like soccer and 5 like swimming. 3 of them like both. How many kids like at least one of the two sports? (*Use what you noticed in problem 2.*)

4. The same club of 10 kids as in problem 3. How many of them like neither soccer nor swimming?

5. In a group of 12 kids, 7 like cookies and 6 like cake. 11 of them like at least one of the two. How many kids like **both**?

6. In a class of 25 students, 15 like pizza, 10 like ice cream, and 6 like both. How many students like neither pizza nor ice cream?

Answer Key — for parents

1. Doing one of them: $8 + 5 = 13$ kids (since no one is on both, simple addition works). Doing neither: $20 - 13 = 7$ kids.
2. (a) **Ana** and **Beto** appear on both lists. (b) The friends who like dogs or cats (or both) are Ana, Beto, Carla, Diego, Elena — all **5**. (c) **No**. $3 + 4 = 7$, but the answer is 5. Ana and Beto each got counted twice (once as dog-lovers, once as cat-lovers). The double-counted amount is 2 — exactly the number who like both. (*This is the key insight: $7 - 2 = 5$.*)
3. Adding $6 + 5 = 11$ double-counts the 3 kids who like both, so subtract them: $6 + 5 - 3 = 8$ kids like at least one sport.
4. Total kids minus those who like at least one: $10 - 8 = 2$ kids like neither.
5. $7 + 6 = 13$, but only 11 kids like at least one. The 2 extras are kids counted twice — the ones who like both. So **2** kids like both. (*Same rule, rearranged: $\text{both} = \text{liked-cookies} + \text{liked-cake} - \text{liked-at-least-one}$.*)
6. Liking pizza or ice cream (or both): $15 + 10 - 6 = 19$. Liking neither: $25 - 19 = 6$ students. (*This is the same shape as Saturday Test No. 4, problem 6.*)